

Improved image quality and reduced acquisition time in brain MRI using deep learning-based reconstruction: A quantitative and subjective assessment compared to standard MPRAGE in 0.55 T MRI

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ABSTRACT

Purpose: To assess the impact of deep learning (DL)-based image reconstruction on quantitative and subjective image quality in brain MRI at 0.55 T by comparing DL-reconstructed MPRAGE with standard MPRAGE using comparable acquisition geometry and timing parameters.

Methods: In this prospective study, 30 patients underwent two consecutive 3D T1-weighted MPRAGE acquisitions on a 0.55 T MRI system (MAGNETOM Free.Max, Siemens Healthineers): a standard reconstruction and a DL-based reconstruction generated from undersampled k-space data using a variational-network architecture. Identical circular regions of interest (ROIs) were placed in gray matter (caudate nucleus), white matter (centrum semiovale), cerebrospinal fluid (lateral ventricle), and air for noise estimation. Signal-to-noise ratio (SNR) and contrast-to-noise ratio (CNR) were computed. Three blinded radiologists independently evaluated image quality using a 5-point Likert scale based on anatomical detail, gray–white matter differentiation, vascular visibility, artifact burden, and overall diagnostic quality.

Results: DL-based reconstruction significantly improved objective image quality across all metrics. SNR increased from 40.83 ± 10.11 to 106.30 ± 51.38 in gray matter ($p < 0.0001$), from 48.95 ± 12.52 to 125.30 ± 58.01 in white matter ($p < 0.0001$), and from 5.77 ± 1.10 to 7.75 ± 4.76 in CSF ($p = 0.021$). CNR between gray and white matter increased from 8.12 ± 4.24 to 18.99 ± 11.21 ($p < 0.0001$).

Subjective ratings favored DL reconstruction for anatomical detail (4.05 ± 0.62 vs. 3.68 ± 0.64), vascular visibility (4.37 ± 0.56 vs. 3.63 ± 0.72), and overall image quality (4.07 ± 0.72 vs. 3.57 ± 0.73). Standard reconstruction showed slightly better artifact suppression (3.80 ± 0.61 vs. 3.27 ± 0.78) and gray–white matter contrast (4.00 ± 0.64 vs. 3.87 ± 0.71).

Acquisition time was reduced from 6:44 min (standard MPRAGE) to 3:06 min with DL reconstruction, corresponding to a substantial scan time reduction.

Conclusion: DL-based reconstruction markedly enhances quantitative and subjective image quality in low-field brain MRI and enables a markedly shorter acquisition time compared to the standard protocol. This study adds clinical evidence that DL reconstruction can substantially improve 0.55 T MPRAGE imaging, supporting its integration into routine neuroimaging workflows to improve efficiency, patient comfort, and diagnostic confidence.

Abbreviations: AI, Artificial Intelligence; AJR, American Journal of Roentgenology; CIs, Confidence Intervals; CNN, Convolutional Neural Network; CNR, Contrast-to-Noise Ratio; CS, Compressed Sensing; DL, Deep Learning; DLR, Deep Learning Reconstruction; ESPIRiT, Eigenvalue-based Parallel Imaging Reconstruction Technique; FLAIR, Fluid-Attenuated Inversion Recovery; MRI, Magnetic Resonance Imaging; MPRAGE, Magnetization-Prepared Rapid Gradient Echo; PI, Parallel Imaging; ROI, Region of Interest; SD, Standard Deviation; SE, Standard Error; SNR, Signal-to-Noise Ratio; T1, Longitudinal Relaxation Time; T2, Transverse Relaxation Time; TI, Inversion Time; TE, Echo Time; TR, Repetition Time.

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Key Points:

- **Question:** Low-field brain MRI provides limited image quality and long acquisition times; improved reconstruction techniques are required to enhance diagnostic confidence and clinical workflow efficiency.
- **Findings:** DL-based reconstruction significantly increased SNR and CNR, improved radiologist-rated image quality, and reduced MPAGE scan time substantially at 0.55 T.
- **Clinical Relevance Statement:** By improving image quality and markedly reducing acquisition time, DL-based reconstruction enhances patient comfort, reduces motion artifacts, and enables efficient, high-quality low-field neuroimaging in routine clinical practice.

1. Introduction

Magnetic resonance imaging (MRI) plays a pivotal role in neuroimaging due to its superior soft tissue contrast, lack of ionizing radiation, and multiparametric capabilities. In recent years, the increasing demand for faster acquisition times and higher image quality has driven the development of advanced reconstruction algorithms. Among them, deep learning-based methods such as Deep Resolve (Siemens Healthineers, Forchheim, Germany) have emerged as promising tools to enhance image sharpness and reduce acquisition time without compromising diagnostic accuracy [1–3].

Deep learning (DL) reconstruction approaches utilize convolutional neural networks (CNNs) trained on large datasets to learn the statistical distribution of noise. These models enable image reconstruction from undersampled raw data, thereby improving efficiency while maintaining or enhancing image quality [4–6]. Compared to traditional methods such as parallel imaging (PI) or compressed sensing (CS), DL-based reconstructions are less prone to common artifacts like noise amplification or residual aliasing and offer improved preservation of anatomical edges and tissue contrast [7,8].

T1-weighted 3D MPAGE (Magnetization-Prepared Rapid Gradient Echo) sequences are routinely used in brain imaging and form the backbone of many clinical and research protocols, especially in volumetric morphometry, lesion detection, and pre-surgical planning. High spatial resolution and clear differentiation between gray and white matter are essential for diagnostic accuracy. Therefore, any improvement in the sharpness and contrast of MPAGE images has immediate clinical value, particularly in patients with subtle neurodegenerative, vascular, or inflammatory pathologies [9,10].

Despite the theoretical advantages of DL-based reconstruction, prior studies have frequently been conducted in phantom experiments or controlled research settings, although an increasing number of clinical investigations have reported promising results. To further support clinical translation, additional studies evaluating AI-assisted reconstruction in routine clinical practice remain warranted, combining objective image quality metrics such as signal-to-noise ratio (SNR), contrast-to-noise ratio (CNR), and edge sharpness with subjective assessment by expert radiologists [11,12].

The present study aims to fill this gap by comparing standard MPAGE sequences with and without DL reconstruction in a clinical brain MRI setting. Using a cohort of 30 patients scanned under identical imaging parameters, we performed intra-individual comparisons with consistent ROI placement. The evaluation included quantitative image quality metrics (SNR and CNR) as well as subjective image quality assessment by three blinded readers. We hypothesized that Deep Resolve would yield superior image quality in terms of signal, contrast, and structural sharpness, while simultaneously reducing acquisition time, thus improving patient throughput and scanner efficiency.

2. Materials and methods

2.1. Study design and patient population

This prospective, single-center study was conducted to evaluate the impact of deep learning-based image reconstruction on image quality in routine brain MRI. Thirty adult patients (mean age: 66 ± 13.6 years; 18 males, 12 females) undergoing clinically indicated MRI of the brain were prospectively enrolled between September 3, 2024, and November 22, 2024. All participants provided written informed consent prior to imaging. Patients were included irrespective of their underlying neurological diagnosis, as the study aimed to assess technical image quality using paired within-subject comparisons rather than disease-specific diagnostic performance. The study was approved by the institutional ethics committee.

2.2. MRI acquisition protocol

MRI was performed using a 0.55 T scanner (MAGNETOM Free.Max, Siemens Healthineers, Forchheim, Germany). Each patient underwent two consecutive acquisitions during a single session. The first scan was the routinely used 3D sagittal T1-weighted MPAGE sequence using a standard parallel imaging reconstruction. For the second scan a research sequence with DL-based reconstruction was employed. The latter generates images from undersampled k-space data using an architecture inspired by variational networks [9,10] that was previously explored for dynamic liver imaging [9,10]. The algorithm performs six iterations that alternate between parallel imaging-based data consistency and neural network-based image enhancement. The model parameters were determined through supervised training in an end-to-end manner using about 1000 fully sampled datasets obtained from healthy volunteers in different body regions utilizing routine 1.5 T and 3 T scanners (MAGNETOM scanners, Siemens Healthineers, Forchheim, Germany). The obtained model parameters were then integrated into the reconstruction pipeline of the scanner for prospective use.

Compared with the standard MPAGE, the DL-based sequence was reconstructed with a higher effective in-plane resolution, while slice thickness and timing parameters (TR, TE, TI) were kept constant. For both standard and DL-based MPAGE, the acquisition matrix and acquisition geometry (orientation and coverage) were identical (Table 1). Differences between protocols arose from the vendor-integrated DL reconstruction workflow, which incorporates a DL-based interpolation and upsampling component. As a result, the reconstruction matrix and effective in-plane spatial resolution differed between the standard and DL-based reconstructions, despite identical acquisition settings. This implementation is in line with previously described two-step DL reconstruction approaches, combining undersampled acquisition with DL-based reconstruction and upsampling [13].

2.3. ROI placement and quantitative analysis

For quantitative image quality assessment, circular regions of

Table 1

Imaging parameters of the standard and deep learning-based MPRAGE sequences at 0.55 T. Acquisition settings were nearly identical to ensure direct comparability, except for a higher parallel imaging acceleration in the DL.

Parameter	Standard MPRAGE	DL-MPRAGE
Field strength	0.55 T	0.55 T
Sequence type	3D MPRAGE	3D MPRAGE (DL-reconstructed)
Orientation	Sagittal	Sagittal
Repetition time (TR)	2200 ms	2200 ms
Echo time (TE)	5.16 ms	5.16 ms
Inversion time (TI)	800 ms	800 ms
Flip angle	11°	11°
Field of view (FOV)	250 × 250 mm	250 × 250 mm
Acquisition matrix (base × phase)	224 × 224	224 × 224
Reconstruction Matrix	224 × 224	448 × 448
Phase resolution	100%	100%
Slice resolution	81%	81%
Voxel size (mm ³)	1.1 × 1.1 × 1.0	0.6 × 0.6 × 1.0
Slice thickness	1.0 mm	1.0 mm
Number of slices	192	192
Acceleration technique	CAIPIRINHA	CAIPIRINHA + DL
Acceleration factor (PE × 3D)	2 × 1	3 × 1
Partial Fourier (slice)	6/8	6/8
Interpolation	Off	On
Bandwidth	130 Hz/pixel	130 Hz/pixel
DL reconstruction	–	DL-CAIPI-6
Denosing strength	–	0.2
Acquisition time	6:13 min	3:06 min

interest (ROIs) were manually placed on matched sagittal slices using consistent anatomical landmarks. The same ROIs were used for both standard and DL-reconstructed images within each patient. Both MPRAGE acquisitions were performed consecutively within the same examination session using identical slice orientation and coverage. ROI placement was performed on the Syngo.via workstation (Siemens Healthineers), which allows direct copy-and-paste of ROIs between image datasets. Identical ROIs (same size, shape, and location) were transferred between standard and DL-reconstructed images and visually verified on anatomically matched slices, ensuring accurate intra-individual comparability without the need for additional image registration.

The following ROI positions were defined (Fig. 1):

- ROI 1 – Gray Matter: Placed in the head of the caudate nucleus
- ROI 2 – White Matter: Placed in the centrum semiovale
- ROI 3 – CSF: Placed in the lateral ventricle
- Noise ROI: Placed outside the skull in air (~3 cm from the head), avoiding any anatomical structures

Each ROI was manually drawn with the same size across all measurements and was identically placed in both the standard and DL-based reconstructions for each patient to ensure consistent and comparable analysis.

Following quantitative metrics were calculated to evaluate image quality:

1. Signal-to-noise ratio (SNR) is defined as the ratio of the mean signal intensities of different ROIs to the standard deviation of the noise:

$$SNR = \frac{\text{mean signal intensities}}{\text{standard deviation of noise}}$$

2. Contrast-to-noise ratio (CNR) is defined as the difference in mean signal intensity ROIs divided by the standard deviation of the background noise:

$$CNR = \frac{\text{difference in signal intensities}}{\text{standard deviation of noise}}$$

2.4. Subjective image quality assessment

Following the MRI acquisition, all image datasets were transferred to the local Picture Archiving and Communication System (PACS) and the Syngo.via workstation (Siemens Healthineers, Forchheim, Germany) for clinical review. A standardized 5-point Likert scoring system was employed to qualitatively assess the images based on predefined criteria including the visual contrast between gray and white matter, the clarity and delineation of anatomical structures, and the visibility of fine vascular details. The presence and severity of image artifacts, as well as the overall diagnostic quality of the image.

Each parameter was rated independently by three radiologists: one with over 10 years of experience, one with 5 years of experience, and one with 2 years of experience. All radiologists were blinded to the reconstruction technique (standard vs. DL). Scores ranged from 1 (poor) to 5 (excellent) for each criterion.

Inter-observer agreement for overall image quality was formally assessed using the intraclass correlation coefficient (ICC), as described in the Statistical Analysis section.

2.5. Statistical analysis

Image quality metrics included signal-to-noise ratio (SNR) in gray matter, white matter, and cerebrospinal fluid (CSF), as well as contrast-to-noise ratio (CNR) between gray and white matter. Only cases with complete paired measurements for both the standard and the DL-based reconstruction were included in the analysis. All 30 patients met this criterion and were included.

For each metric, descriptive statistics (mean ± standard deviation [SD]) were calculated separately for both reconstruction techniques. Paired differences (DL – Standard) were assessed for normality using the Shapiro–Wilk test. If normality was satisfied, paired two-sided Student's *t*-tests were performed; otherwise, the Wilcoxon signed-rank test was applied. Effect sizes for paired data were expressed as Cohen's *d*. In addition, 95% confidence intervals (CIs) of the mean differences were computed.

To account for multiple comparisons across the four primary metrics, *p*-values from the paired *t*-tests and Wilcoxon tests were adjusted using the Holm–Bonferroni method. A *p*-value <0.05 (after adjustment) was considered statistically significant.

Data visualization included paired line plots, boxplots, histograms of differences, mean ± standard error (SE) plots, and Bland–Altman plots to illustrate bias and limits of agreement between the two reconstruction techniques. All statistical analyses were performed using R (version 4.5.1, 2025.05.1 + 513; R Foundation for Statistical Computing, Vienna, Austria). Statistical tables and visualizations were generated using the dplyr, ggplot2, and openxlsx packages in R.

Inter-observer agreement among the three readers for overall image quality was assessed using the intraclass correlation coefficient (ICC). A two-way random-effects model with absolute agreement for single measurements (ICC [1,2]) was used, as recommended for reliability studies involving multiple independent raters.

3. Results

3.1. Objective image quality metrics

Quantitative evaluation showed a marked improvement in image quality metrics with DL-based reconstruction compared to the standard MPRAGE sequence.

The signal-to-noise ratio (SNR) in gray matter increased from 40.83 ± 10.11 to 106.30 ± 51.38 (*p* < 0.0001), and in white matter from 48.95 ± 12.52 to 125.30 ± 58.01 (*p* < 0.0001). Similarly, SNR in cerebrospinal fluid (CSF) improved from 5.77 ± 1.10 to 7.75 ± 4.76 (*p* = 0.021).



Fig. 1. ROI placement for quantitative evaluation: ROI 1 in the head of the caudate nucleus (gray matter), ROI 2 in the centrum semiovale (white matter), ROI 3 in the lateral ventricle (CSF), and the noise ROI placed in air outside the skull (~3 cm from the head).

The contrast-to-noise ratio (CNR) between gray and white matter also increased significantly from 8.12 ± 4.24 to 18.99 ± 11.21 ($p < 0.0001$), reflecting enhanced tissue contrast. Table 2 and Fig. 2 illustrate the quantitative parameter values.

3.2. Subjective image quality assessment

The subjective evaluation of image quality demonstrated significant improvements with DL-based reconstruction compared to standard MPRAGE. Radiologists assessed five quality criteria using a 5-point Likert scale. Mean scores were higher in DL images for delineation of

anatomical details (4.05 ± 0.62 vs. 3.68 ± 0.64), visibility of small vasculature (4.37 ± 0.56 vs. 3.63 ± 0.72), and overall image quality (4.07 ± 0.72 vs. 3.57 ± 0.73). In contrast, gray-white matter contrast showed a slight decrease with DL (3.87 ± 0.71 vs. 4.00 ± 0.64), and artifact level was rated more favorably in the standard sequence (3.80 ± 0.61) compared to DL (3.27 ± 0.78).

These findings are visually summarized in Fig. 3, illustrating the mean ratings across all five parameters. A detailed overview of all scores including standard deviations is presented in Table 3.

Acquisition times were protocol-specific. The DL-reconstructed MPRAGE required 3:06 min, whereas the standard MPRAGE sequence had a duration of 6:44 min. Thus, the DL-based protocol enabled a markedly shorter acquisition time compared with the standard protocol. This substantial time reduction is of high clinical relevance, as shorter scan durations may improve patient comfort, reduce motion-related artifacts, and increase overall scanner throughput in daily routine practice.

Examples comparing standard and DL-based reconstruction are shown in Figs. 4 and 5, illustrating normal brain anatomy and representative pathological findings, respectively.

Inter-observer agreement for overall image quality was limited, with ICC values of 0.37 for standard MPRAGE and 0.14 for DL-MPRAGE, indicating low agreement between readers.

Table 2
Quantitative comparison of image quality between standard and DL-based reconstruction MPRAGE sequences across 30 patients. Mean $\pm$ standard deviation values are shown for signal-to-noise ratio (SNR) in gray matter, white matter, and CSF, as well as contrast-to-noise ratio (CNR) between gray and white matter. DL-based reconstruction resulted in significantly higher values across all parameters ($p < 0.0001$).

Metric	Standard (Mean $\pm$ SD)	DL (Mean $\pm$ SD)	p-value
SNR Gray	40.83 $\pm$ 10.11	106.30 $\pm$ 51.38	< 0.0001
SNR White	48.95 $\pm$ 12.52	125.30 $\pm$ 58.01	< 0.0001
SNR CSF	5.77 $\pm$ 1.10	7.75 $\pm$ 4.76	0.021
CNR Gray-White	8.12 $\pm$ 4.24	18.99 $\pm$ 11.21	< 0.0001

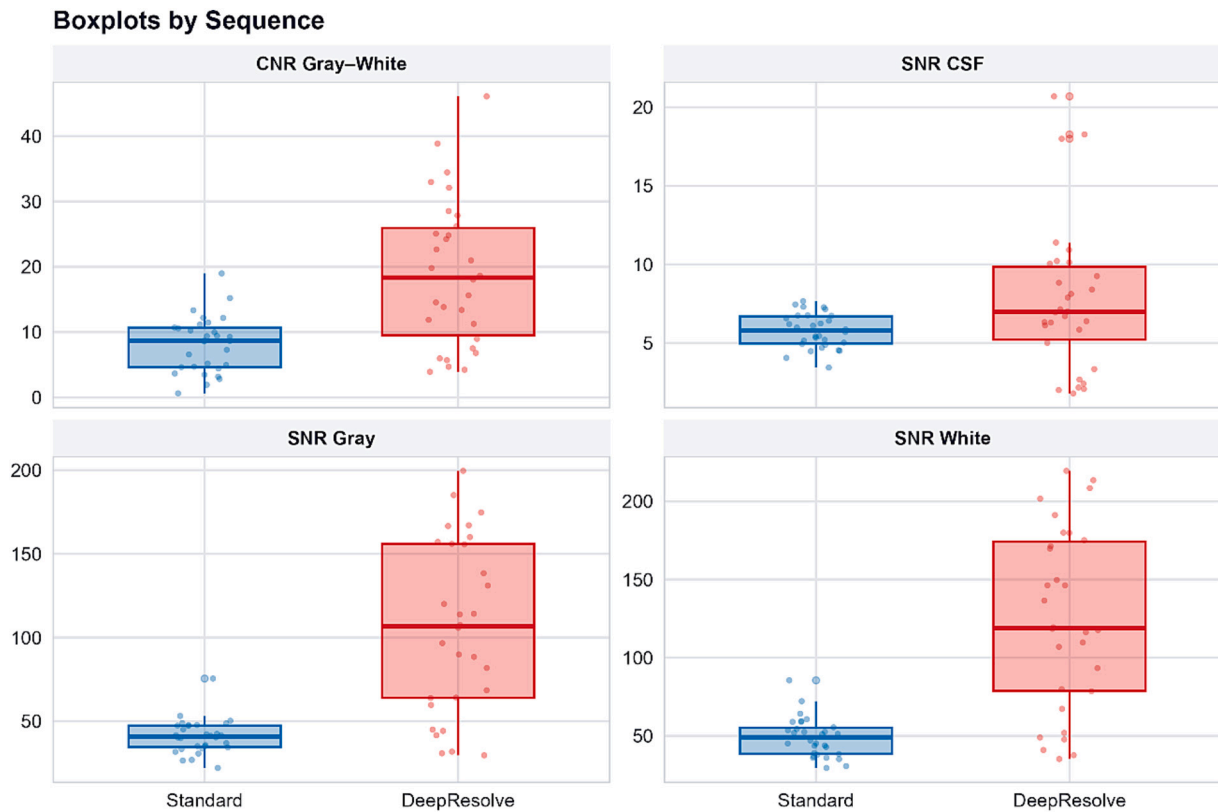


Fig. 2. Boxplots comparing signal-to-noise ratio (SNR) and contrast-to-noise ratio (CNR) between standard MPRAGE and DL-based reconstruction. All metrics (SNR in gray matter, white matter, and CSF, as well as CNR gray-white) show a substantial increase with DL. Each dot represents one patient.

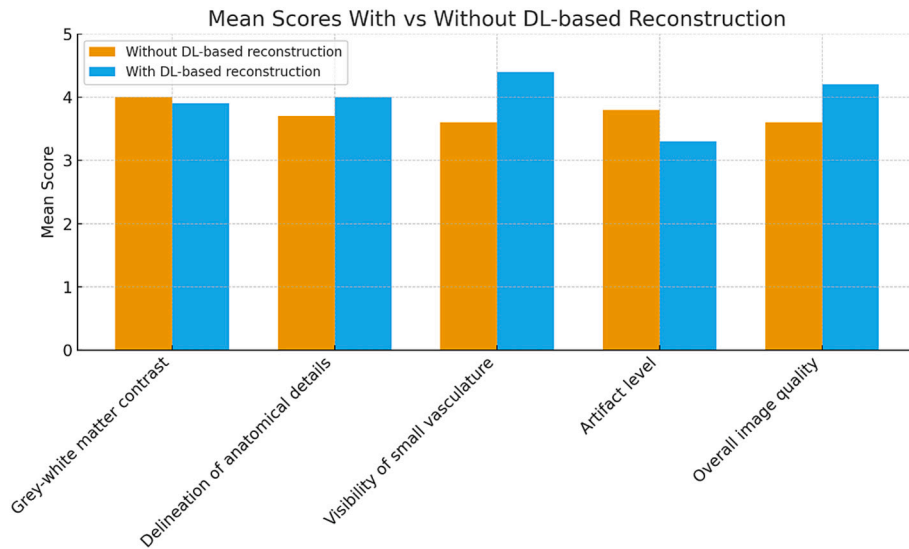


Fig. 3. Bar plot comparing the mean subjective image quality scores for MPRAGE sequences acquired with and without Deep Resolve. Ratings were assigned on a 5-point Likert scale (1 = poor, 5 = excellent). Scores for anatomical detail, vascular visibility, and overall image quality improved with Deep Resolve, while artifact level and gray-white contrast were rated slightly better in the standard sequence.

4. Discussion

This prospective study demonstrates that deep learning-based image reconstruction significantly improves both objective and subjective image quality in brain MRI when compared to standard MPRAGE sequences. The findings reveal a marked increase in signal-to-noise ratio (SNR) and contrast-to-noise ratio (CNR), indicating enhanced tissue differentiation and improved noise suppression, critical factors in

neuroimaging where small structural changes can be diagnostically relevant. These improvements were achieved despite reduced acquisition times, which highlights the potential of DL-based reconstruction to simultaneously optimize image quality and efficiency.

From a technical standpoint, the increased SNR in both gray and white matter may be attributed to the underlying convolutional neural networks (CNN), which are trained to suppress noise while preserving anatomical detail. These findings align with recent reports that

Table 3

Mean subjective image quality ratings ($\pm$ standard deviation) for MPRAGE sequences acquired with and without deep learning-based reconstruction. Scores were assigned on a 5-point Likert scale (1 = poor, 5 = excellent) by three independent radiologists.

Parameter	conventional	DL-based reconstruction
gray-white matter contrast	4.00 $\pm$ 0.64	3.87 $\pm$ 0.71
delineation of anatomical details	3.68 $\pm$ 0.64	4.05 $\pm$ 0.62
visibility of small vasculature	3.63 $\pm$ 0.72	4.37 $\pm$ 0.56
artifact level	3.80 $\pm$ 0.61	3.27 $\pm$ 0.78
overall image quality	3.57 $\pm$ 0.73	4.07 $\pm$ 0.72

demonstrated similar gains in accelerated MRI across other body regions, such as the liver, knee, and spine [14–17]. Moreover, our results extend prior work by showing that such improvements can also be achieved at a low field strength (0.55 T), a setting often considered less favorable for high-resolution neuroimaging.

Our findings show that even under these technical constraints, AI-based reconstruction can achieve substantial improvements in both image quality and acquisition time. This supports recent evidence suggesting that advanced AI algorithms may help expand the clinical utility of low-field MRI systems [18,19].

Our results are consistent with recent research on AI-supported reconstruction across various anatomical regions. For example, studies in musculoskeletal and abdominal MRI have shown that DL-based reconstruction significantly enhances edge sharpness and overall image quality, while also shortening scan times [20,21]. However, these studies were largely performed at standard field strengths (1.5 T or 3 T).

While previous reports have demonstrated general improvements in image quality using AI reconstruction for brain MRI [16], recent work by Bieri et al. [22] has reported an optimized 0.55 T brain MP-RAGE protocol incorporating deep learning-based reconstruction with scan-time reduction. In this context, our study provides a prospective clinical evaluation of DL-reconstructed MPRAGE at 0.55 T using paired intra-individual acquisitions with identical geometry, combining quantitative SNR/CNR analysis and blinded multi-reader qualitative assessment.

The subjective image evaluation was in line with the objective findings. On average, DL-reconstructed images received higher ratings for anatomical detail, vascular visibility, and overall image quality. These aspects are crucial in clinical neuroimaging, where subtle anatomical features guide diagnostic decisions [16].

The combined advantage of improved image quality and shorter scan time has direct clinical relevance. Faster scans reduce the risk of motion artifacts and improve patient compliance, particularly in vulnerable populations such as pediatric or elderly patients and contribute to more efficient use of scanner resources in busy radiology departments.

Although DL-based reconstruction improved objective image quality metrics and overall subjective image quality, DL images were rated slightly inferior with respect to artifact level and gray–white matter contrast. This may be explained by the pronounced denoising characteristics of DL reconstruction, which can lead to smoother image texture and subtle alterations in edge appearance, thereby affecting perceived contrast despite objectively improved SNR and CNR. Similar observations have been reported in prior studies on AI-based MRI reconstruction and highlight the difference between quantitative image metrics and subjective visual perception [17].

Conversely, the markedly shorter acquisition time of the DL-based MPRAGE is expected to reduce motion-related artifacts, which likely contributed to the improved delineation of anatomical details, enhanced visibility of small vasculature, and higher overall image quality ratings. The observed scan time reduction reflects a combination of higher acceleration, vendor-specific sequence implementation, and DL-based reconstruction and interpolation, rather than acceleration factor alone.

While objective CNR increased with DL reconstruction due to effective noise suppression, subjective assessment of gray–white matter contrast was slightly lower. This highlights that CNR reflects signal

differences relative to noise, whereas perceived contrast also depends on edge definition and image texture, which may be altered by DL-based denoising. In this context, both acquisition-related factors and reconstruction-related effects may interact and jointly influence perceived tissue contrast and artifact appearance.

Moreover, the pronounced denoising effect of DL-based reconstruction may result in very low noise standard deviations, which could amplify relative signal variations and contribute to the wider distribution of SNR and CNR values observed in the DL images.

In addition, the slightly reduced gray–white matter contrast and differences in artifact levels observed in the DL-based images may be influenced by several interacting factors. Differences in k-space sampling order following the inversion pulse, particularly the timing of central k-space acquisition relative to the inversion time, as well as higher acceleration factors and DL-based reconstruction processes, may affect image texture and perceived tissue contrast. At the same time, the markedly shorter acquisition time of the DL-MPRAGE protocol is expected to reduce motion-related artifacts, which may partially compensate for these effects. These reconstruction- and acquisition-related factors should be considered when interpreting subjective image quality differences between both protocols.

Although the exact sampling strategies of the vendor-integrated DL reconstruction pipeline are not fully disclosed, such differences may have contributed to the slightly reduced contrast observed in the DL-based images. This aspect represents a potential confounding factor and should be considered when interpreting the present results. Future studies with detailed access to acquisition schemes are warranted to further investigate this effect.

However, this study also has limitations. It was conducted at a single center using a single low-field MRI system (0.55 T), which may limit the generalizability of the results to other field strengths or vendors. While both objective (SNR/CNR) and subjective (radiologist rating) assessments were performed, the diagnostic accuracy in specific clinical scenarios was not evaluated. The study focused exclusively on the MPRAGE sequence; the performance of DL-based reconstruction in other pulse sequences (e.g., FLAIR, T2-weighted) remains to be explored in future research. Disease-specific effects on image quality were not separately analyzed, as each patient served as their own control in the paired comparison design.

Another limitation of the present study is the relatively low inter-observer agreement for subjective image quality ratings. This finding highlights the subjective nature of visual image quality assessment and supports the use of quantitative metrics in future studies.

Dedicated morphometric and volumetric validation using standardized neuro-morphometric pipelines (e.g., FreeSurfer) was beyond the scope of the present study and should be addressed in future, specifically designed investigations.

In summary, our study highlights the robust potential of DL-based reconstruction to elevate low-field brain MRI by improving image quality and reducing scan times. These findings support broader adoption of AI-based reconstruction techniques as a key tool in modern clinical radiology.

Statistics and biometry

No complex statistical methods were necessary for this paper.

Study subjects or cohorts overlap

No study subjects or cohorts have been previously reported.

Methodology

Prospective diagnostic study performed at one institution.



Fig. 4. Example of normal anatomy in MPRAGE: standard reconstruction (left) and DL-based reconstruction (right) shown in the same slice and plane, enabling direct comparison of image quality.

Originality of the work

This manuscript is an original work that has not been published previously and is not under consideration for publication elsewhere, in

whole or in part.



Fig. 5. Example with pathological findings: the upper row shows mucosal swelling of the paranasal sinuses, and the lower row demonstrates an osseous metastasis in the same slice and plane. DL-based reconstruction provides improved delineation of lesion margins, better overall image quality, and clearer distinction of pathological structures compared to the standard reconstruction.

Exclusivity of submission

The manuscript is submitted exclusively to *Magnetic Resonance Imaging* and will not be submitted to any other journal while under review.

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Corresponding author responsibilities

Dr. Saher Saeed, as the corresponding author, accepts responsibility for all communication with the journal and will ensure that all co-authors are kept fully informed during the submission, review, and publication process.

CRediT authorship contribution statement

Saher Saeed: Writing – original draft, Supervision, Project administration, Methodology, Formal analysis. **Dominik Nickel:** Methodology. **Elisabeth Weiland:** Methodology. **Lynn Johann Frohwein:** Methodology. **Julius Henning Niehoff:** Data curation. **Georg Reiman:** Software. **Julian Schreck:** Writing – original draft. **Nina Haag:** Conceptualization. **Jan Boriesosdick:** Supervision. **Denise**

Schoenbeck: Software. **Maria Katz:** Project administration. **Iram Shahzadi:** Software. **Matthias Michael Woeltjen:** Methodology. **Jan Robert Kroeger:** Writing – review & editing, Supervision. **Jan Borggreffe:** Writing – review & editing, Supervision. **Christoph Moeninghoff:** Writing – review & editing, Supervision.

Informed consent

Written informed consent was obtained from all subjects (patients) in this study.

Ethical approval

Institutional Review Board approval was obtained. The study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Ethics Committee of the Faculty of Medicine of the Ruhr University Bochum, Germany.

Author agreement statement

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Declaration of competing interest

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Data availability

The authors confirm that all data are accurate, complete, and honestly reported. No data fabrication, falsification, or inappropriate manipulation has occurred.

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